Response to Reviewer 1

Joint Evidence Gating for Continual Maintenance of Predictive Digital Twin Models Under Unreliable Monitoring

Computers, Materials & Continua

Dear Reviewer 1,

Thank you for your comments. Below we answer each point, summarize the relevant evidence and identify its location in the revised Clean manuscript. Supplement references refer to the accompanying Supplementary Protocol and Results.

R1.1

Comment. Section 3 frames RAVM-DT as a general algorithm for "a DT platform that maintains center-specific models for a set of heterogeneous computing centers." However, all experimental evaluation — including the Alibaba Cluster Trace in Section 5.7 — addresses the same task: workload (resource utilization) prediction for computing centers. The generality claim is therefore unsupported by the evidence. Digital twin maintenance problems arise in power grids, manufacturing processes, autonomous vehicles, and other domains with qualitatively different data characteristics, drift patterns, and corruption modes. Without at least one additional DT task domain, the reader cannot determine whether RAVM-DT's safety–utility trade-off is domain-specific or genuinely general.

Response. The study concerns predictive DT model maintenance, with workload and measured temperature forecasting as distinct task families. The added thermal evaluation uses three UCI room targets from one home and two ThermBuild houses at one site. Every tested thermal stream fails the transferred source risk criterion. On the 40 previously exposed Alibaba machines, Joint meets the empirical risk criterion but loses the relative-utility comparison with DualMean. These results support the narrower predictive-maintenance scope and are retained in the abstract, results and discussion.

Changes in the revised manuscript. Section 1 (p. 1); Section 5.1.1 (p. 6); Section 5.2.6 (p. 13); Table 7 (p. 14); Section 5.3 (p. 15).

R1.2

Comment. Figure 3(d) shows that the Validation-Commit (Val.-Commit) baseline achieves a 0.0% commit rate across all five primary conditions, it never deploys any candidate throughout the evaluation. As a result, Val.-Commit is functionally identical to Frozen in MSE and harmful transitions (both at 4.843×10^{-3} and 0.0% harmful transitions, respectively).

Response. The original ValidationCommit baseline uses a calibrated retain-all fallback and consequently commits zero candidates. None of its five tested thresholds met the mixed-stream harm criterion; even the largest yielded 123 harmful commitments in 1,440 opportunities (8.54%). At threshold zero, the original clean/mixed calibration committed 869/1,440 and 1,054/1,440 candidates, including 142 and 363 harmful commits. Supplement Table S2 retains that evidence separately from Main. The revised comparison includes active RecentMean, MWU-Mean and DualMean controls, and reports the candidate-generation cost of inactive ValidationCommit.

Changes in the revised manuscript. Section 5.1.2 (p. 7); Table 2 (p. 9); Section 5.1.3 (p. 8); Table 8 (p. 15). Supplement S2, Table S2.